

Yumin Kang

🏢 Human-Centered Intelligent Systems Lab. (HCIS Lab.)

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EDUCATION

Ph.D. in Intelligent Robotics

Gwangju Institute of Science and Technology (GIST)

Advisor: Prof. SeungJun Kim

Mar 2024 – Present

Gwangju, Korea

M.S. in Intelligent Robotics

Gwangju Institute of Science and Technology (GIST)

Advisor: Prof. SeungJun Kim

Mar 2022 – Feb 2024

Gwangju, Korea

B.S. in Media Technology

(AI & Data Analytics / IOT Service / Design Engineering)

Hanyang University ERICA

Advisor: Prof. Gyouhyung Kyung

Mar 2018 – Feb 2022

Ansan, Korea

RESEARCH EXPERIENCE

Research Assistant

Human-Centered Intelligent Systems Laboratory, GIST

Advisor: Prof. SeungJun Kim

Mar 2024 – Present

- SpaceTop: Spatial Computing HCI Technology for Everywhere XR Productivity Workstation, IITP
- Actuated XR Systems Powered by Sensory Intelligence and Soft Robotics, NRF
- Development of hyper-connected mobility convergence technology based on AI to build next-generation smart city, KAIA
- HCI + AI for Human-Centered Physical System Design, GIST-MIT Joint Research
- Inter-university Alliance for Future vehicular Technology, KIAT

Research Assistant

Human-Centered Intelligent Systems Laboratory, GIST

Advisor: Prof. SeungJun Kim

Mar 2022 – Feb 2024

- Meta-Human with Immersive Mobility Platform and Multimodal XR Contents, RAPA
- Inter-university Alliance for Future vehicular Technology, KIAT
- HCI + AI for Human-Centered Physical System Design, GIST-MIT Joint Research
- GIST Metaverse Research Center: Metaverse with Immersive Transportation Methods – Walking and Driving XR for Meta-mobility, RAPA

Research Intern

Jul 2021 – Feb 2022

Human-Centered Intelligent Systems Laboratory, GIST

Advisor: Prof. SeungJun Kim

- GIST CarXR Lab: In-car XR platforms and contents to augment passenger UX in futuristic vehicles, RAPA

Research Intern

Apr 2020 – Jun 2021

IXLAB (Interaction & eXperience Laboratory), Hanyang University

Advisor: Prof. Gyouhyung Kyung

- Ergonomic evaluation and application of horizontally, vertically, and bi-directionally curved displays considering 2D/3D horopters and static and dynamic contexts of display use, NRF

PUBLICATIONS

- Um, J., Gim, B., Kang, S., **Kang, Y.**, Jeon, E., Kim, S. (2025). TeleHopper: Simulating a Jumping Sensation as Proprioceptive Feedback for Teleportation in Virtual Reality via Electrical Muscle Stimulation. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems
- Um, J., Jeon, E., **Kang, Y.**, Kang, S., Elsharkawy, A., DelPreto, J., Matusik, W., Rus., D., Kim, S. (2024). LegSense: Inducing Walking Sensation in Seated VR by Providing Movement Illusion via Electrical Muscle Stimulation. In Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing.
- Jeon, H.* , Jo, T.* , Yeo, D.* , An, E., **Kang, Y.**, Kim, S. (2024). The Way of Water: Exploring the Role of Interaction Elements in Usability Challenges with In-Car VR Experience. Virtual Reality 28, 121.
- Seong, M., Kim, G., **Kang, Y.**, Jang, J., DelPreto, J., Kim, S. (2024). Counterfactual Explanation-Based Badminton Motion Guidance Generation Using Wearable Sensors. 2024 IEEE ICRA Workshop on Wearable.
- Seong, M., Kim, G., Yeo, D., **Kang, Y.**, Yang, H., DelPreto, J., Matusik., W., Rus., D., Kim, S. (2024). MultiSenseBadminton: Wearable Sensor–Based Biomechanical Dataset for Evaluation of Badminton Performance. Scientific Data 11, 343.
- **Kang, Y.**, Choi, S., An, E., Hwang, S., Kim, S. (2023). Designing Virtual Agent Human–Machine Interfaces Depending on the Communication and Anthropomorphism Levels in Augmented Reality. In Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications.

AWARDS

- Honorable Mention**
AutomotiveUI 2023

Sep 2023
- ERICA Software-Up Makeathon 4th prize**
Hanyang University

Feb 2019

SCHOLARSHIPS

- GIST Scholarship (Government supported)**
4 years, Full funding, doctoral degree

Mar 2024 – Present
- GIST Scholarship (Government supported)**
2 years, Full funding, master’s degree

Mar 2022 – Feb 2024
- Scholarship of Academic Excellence**
Hanyang University ERICA

Spring 2020

TEACHING EXPERIENCE

- Teaching Assistant**

RT5301/AI6316/FE5403 *Human-Computer Interaction*

RT5302/AI6317/CT5301/FE5404 *RM in HCI*

Spring 2023,
Spring 2025

Fall 2024
- Guest Lectures**

“Human–Vehicle Interaction: Investigating Communication Strategies Between Automated Vehicles and Road Users,” *Human–Computer Interaction*

“Working with Human Subjects,” *RM in HCI*

Spring 2025

Fall 2024

ACADEMIC SERVICE

- Organizing Committee**

- Organizer**
Ubicomp/IWSC 2024 Workshop on *Interpretable, Inclusive, and Immersive Interaction for Ubiquitous AI-infused Physical Systems*

- **Reviewer**

Special Recognition for Outstanding Review
CHI 2025, CHI 2025 LBW, IMWUT 2025

Conferences

CHI 2025, DIS 2025, AutomotiveUI 2024

Journals

IMWUT 2025

Posters

CHI 2025 LBW

LANGUAGES

- **Korean** (Native)
- **English** (Advanced working proficiency / TOEIC 875, TOEIC SPEAKING 150)